

Optimization as Likelihood

A Bayesian Review of Auyon Siddiq’s Research on Inverse Optimization, Incentives, Platforms and AI

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ABSTRACT. Auyon Siddiq’s research treats observed decisions as solutions of optimization problems, recovers their parameters from data, and designs incentives, infrastructure or platform rules on top of the estimates. The common object is a likelihood with an argmin inside it, and writing it down changes both theory and practice. For inverse optimization with noisy data our main findings are these. In the linear case inverse optimization is a change point problem: least squares converges at rate n^{-1} to the argmax of a compound Poisson process and is inefficient, while the posterior mean is efficient and its credible intervals are calibrated. In the strictly convex case it is a regular nonlinear regression, so classical theory applies, and the published Table 1 errors for one design fall below the information bound by up to a factor of nine. When identification fails the posterior recovers the identified set. The inconsistency of the earlier heuristics comes from observations that carry no information yet enter their loss. The thermostat application is a two sided Tobit model, on which least squares is inconsistent for the comfort energy tradeoff under preference noise while a Bayesian Tobit is not. The Fisher information is an explicit functional of the tangent spaces of the active constraints, and the rate exponent is read off the Hellinger geometry of the solution map. In the linear limit experiment, a location experiment, the posterior mean is the Pitman estimator and least squares is not asymptotically minimax. A Gibbs relaxation of the argmin models bounded rationality and, under the paper’s own assumptions, turns the problem into a concave exponential family estimation solvable in randomized polynomial time. For the study of generative AI and labor commoditization on Upwork we prove that the Shapley block design splits a pure level shift in exposed categories equally across the blocks that encode category, which reproduces the headline human capital result under a null of no change in valuation. We propose Bayesian replacements for the estimate then optimize pipelines and solve ten open problems, including the limit experiment E_κ for $nT \rightarrow \kappa$ that interpolates between regular and change point inverse optimization, with small temperature risk $c_0 = 1/2$ to three decimals, the joint limit experiment for mixed regimes, and the inconsistency of profiling over optimal faces; we also show that least squares on $\log(1 + y)$ manufactures a commoditization effect under the null that a Poisson model does not.

1. INTRODUCTION

Siddiq’s work, much of it with Anil Aswani, Max Shen, Terry Taylor, Nur Kaynar, Sheng Liu, Jingwei Zhang, Shreyas Sekar and Niuniu Zhang, is unusually coherent. The template is: (i) posit that observed decisions are solutions of a forward optimization problem with unknown parameters; (ii) estimate those parameters from noisy decisions; (iii) solve a design problem (an incentive contract, a bike lane network, a defibrillator deployment, a platform fee) using the estimates. The flagship paper is *Inverse Optimization with Noisy Data* [8], which introduced statistical consistency into a literature that had relied on slack based heuristics [14, 54]. The template recurs in the Medicare

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³Disclosure. Auyon Siddiq coauthored, with Andrew Gelman, a published reviewer statement concerning AI assisted papers by the present authors. We note this so readers can weigh it; the arguments below concern only the published research and are offered so they can be checked on their merits. We are sending this draft to Professor Siddiq before circulating it and invite his response.

Shared Savings Program study [9], the estimation of agent production from randomized incentive contracts on Mechanical Turk [53], the Chicago bike lane study [57], and, in a different register, the 2026 study of human capital and generative AI on Upwork [80].

Our reading is that step (ii) is always the construction of a likelihood of the form

$$p(y \mid u, \theta) = \int p(y \mid x) \pi(dx \mid u, \theta), \quad \pi(\cdot \mid u, \theta) \text{ supported on } S(u, \theta) = \arg \min_{x \in \Phi(u, \theta)} f(x, u, \theta), \quad (1)$$

where $p(y \mid x)$ is a measurement model and π selects among optimal solutions. Once (1) is written down, four consequences follow, and they organize this review. First, the behavior of any estimator is governed by the smoothness of $\theta \mapsto S(u, \theta)$; when this map jumps, classical asymptotics fail and Bayes estimators beat least squares (Section 2). Second, the argmin can be replaced by a Gibbs measure at temperature T , which is the quantal response model of bounded rationality and makes the likelihood smooth (Section 3). Third, step (iii) should integrate over posterior uncertainty rather than plug in a point estimate (Section 4). Fourth, when the first stage is a flexible prediction model whose outputs are used as outcomes, the attribution scheme itself can generate the effect being measured (Section 6).

Most of the review concerns the inverse optimization paper, the foundation of the program. Section 2 lists seven shortcomings, and Sections 2 and 3 prove nine propositions and report simulations showing that a likelihood based analysis is more accurate, calibrated and often cheaper. The aim is constructive: the empirical settings are well chosen, and most of our proposals extract more from the same data. Section 7 treats the remaining papers briefly, and Section 8 solves the open problems raised by the analysis.

2. INVERSE OPTIMIZATION WITH NOISY DATA

2.1. The estimator and its claims

In Aswani et al. [8] the forward problem is $\min_x \{f(x, u, \theta) : g(x, u, \theta) \leq 0\}$ with solution set $S(u, \theta)$, the data are pairs (u_i, y_i) with $y_i = \xi_i + w_i$, $\xi_i \in S(u_i, \theta_0)$, and the estimator minimizes

$$Q_n(\theta) = \min_{x_i \in S(u_i, \theta)} \frac{1}{n} \sum_{i=1}^n \|y_i - x_i\|^2.$$

The authors prove that Q_n is lower semicontinuous, regularize the optimality constraint to ϵ optimality through a duality based reformulation, and obtain risk and estimation consistency by epiconvergence and a Wald type argument. They show the problem is NP hard by reduction from best rank one approximation of a third order tensor, propose an enumeration algorithm over a δ net of Θ for $p \leq 5$, and a semiparametric algorithm that first denoises y_i with a kernel smoother. The paper describes itself as “the first statistical inference perspective on the inverse optimization problem.” It has over 250 citations and is the reference point for data driven inverse optimization in the review of Chan et al. [24].

The paper deserves credit for three things. It was the first in operations research to show that the slack based heuristics then in use are inconsistent; its criterion, distance in decision space, coincides with the Gaussian likelihood and is the right one under its stated model; and its proofs handle the lower semicontinuity of Q_n carefully. Its theorems hold under its assumptions, and nothing below disputes them. The paper asks the right first question, consistency, and stops there. It has no probability model beyond a mean zero error, no estimate of the noise scale, no standard error, interval or test for θ , no rate of convergence or limiting distribution for $\hat{\theta}$, and no way to compare forward models. Writing down (1) supplies all of these, and in the leading cases the Bayesian computation is cheaper than the paper’s estimator.

2.2. Seven criticisms

1. *The statistical perspective has a long prior history.* Recovering the parameters of an optimizing agent from noisy decisions is the core problem of structural econometrics. The nested fixed point estimator of Rust [75], the conditional choice probability estimators of Hotz and Miller [48] and the MPEC formulation of Su and Judd [82] share the nested structure of Aswani et al. [8] and come with likelihoods, standard errors and specification tests; Bayesian versions are in Imai et al. [50] and Norets [65]. Revealed preference theory [1, 85] answers the noiseless inverse problem and its testable implications. In machine learning, inverse reinforcement learning [64] is inverse optimization over policies, and both its Bayesian version [72] and its maximum entropy version [87] specify an explicit probability model for observed decisions. This literature appears in the paper only in passing (Bajari et al. [10] is listed among applications), and its estimators are not compared. The duality based reformulation, which bounds the primal objective by the Lagrangian dual to enforce a zero gap, is closely related to the strong duality reformulations standard in bilevel programming [34, 35] and to the MPEC device of Su and Judd [82].

2. *The noise model does not fit the paper’s own application.* The paper names three sources of noise (measurement error, bounded rationality, model mismatch) and models all three as additive error on the decision. Only measurement error lives in decision space. Bounded rationality and heterogeneous preferences are objective space errors: the agent solves a perturbed problem exactly, so every observed decision is feasible and optimal for some parameter. This is the random utility model [59]; Chan et al. [25] draw the same distinction. In the thermostat study of Aswani et al. [8], the constraint $x \in [70, 76]$ is set to the observed setpoint limits, so every recorded setpoint is feasible; additive Gaussian noise on the optimal setpoint would routinely produce values outside it. A pure decision space error model is therefore hard to reconcile with the data. If instead the variation in setpoints comes from preferences, the least squares estimator is inconsistent for θ (Section 2.9), and the consistency theorems of Aswani et al. [8], which assume decision space error, do not cover the behavioral parameter. Which error model is correct is an empirical question; a likelihood lets the data answer it, and a fixed loss does not.

3. *Incidental latent decisions and an unestimated scale.* The criterion profiles one latent x_i per observation, the incidental parameter structure of Neyman and Scott [63]. With single valued solution sets this is harmless for θ , but σ^2 is never estimated, so the criterion has no likelihood scale and cannot be converted into intervals. When IC(iii) fails and optimal faces have positive dimension, profiling also picks the point of the face nearest to y_i : the residual sum of squares then underestimates σ^2 , and $\hat{\theta}$ is biased toward parameters with large optimal faces. Integrating over a selection distribution on the face, as in (1), removes both effects.

4. *The asymptotic theory is weak where it matters.* Without strict convexity, risk consistency is proved only for the ϵ regularized risk (their Theorem 2), not the original one. No rates are given for $\hat{\theta}$. The semiparametric algorithm, the only one offered for $p > 5$, rests on their Proposition 9, a uniform denoising bound of order $n^{-1/(18m)}$ with bandwidth $n^{-2/(8m+1)}$, where $m = \dim u$; for $m = 10$ this is $n^{-1/180}$, no guarantee at any realistic sample size. The authors note that rates are open. In the linear case the rate is n^{-1} and least squares is inefficient (Sections 2.4 and 2.5); in the strictly convex case it is $n^{-1/2}$ with known constant (Section 2.6).

5. *No uncertainty in any experiment.* Every table reports point estimation or prediction error averaged over replications; none gives an interval for θ . In the real data study the quantities of interest are the comfort energy tradeoff θ_1 and the preferred offset θ_2 , but the paper reports neither

their values nor their uncertainty, only prediction error and a bootstrap p value against two heuristics it has already shown inconsistent. Individualized incentives, a stated motivation, require knowing how precisely an individual's parameters are measured.

6. No model comparison or checking. The forward model is assumed. When it is wrong (the FOP C and SQR experiments) the paper can only report risk consistency within the wrong class. A likelihood gives marginal likelihoods for comparing forward models, posterior predictive checks for detecting misfit [40], and, under misspecification, posterior concentration at the Kullback Leibler projection [55], which is the precise version of risk consistency. Goodness of fit for inverse optimization, developed separately by Chan et al. [23], follows from the likelihood directly.

7. The benchmarks are weak. The only competitors are slack based heuristics [14, 54] that the paper proves inconsistent. The relevant competitors are the maximum likelihood and Bayes estimators under the stated noise model. The enumeration algorithm computes the former, and in the linear case the latter dominates it (Table 1).

2.3. Why the heuristics fail: zero information observations

The counterexample in Aswani et al. [8] showing that the variational inequality (VIA) and KKT (KKA) heuristics are inconsistent becomes transparent once the likelihood is computed. The forward problem is $\min\{x^2 - (\theta + u)x : x \in [0, 10]\}$, so $S(u, \theta) = \min\{(u + \theta)/2, 10\}$, with $u \in \{0, 20\}$ equally likely, Rademacher noise $w = \pm 1$, $\Theta = [0, 10]$ and $\theta_0 = 10$.

PROPOSITION 1. *In this example the observations with $u = 20$ have likelihood free of θ on Θ , and the posterior under any prior with an atom at θ_0 equals a point mass at θ_0 with probability at least $1 - 2^{1-m}$, where m is the number of observations with $u = 0$. The VIA and KKA limits 718/73 and 12080/1833 reported by Aswani et al. [8] arise entirely from the $u = 20$ observations.*

Proof. For $u = 20$ and $\theta \in [0, 10]$, $(20 + \theta)/2 \geq 10$, so $S(20, \theta) = \{10\}$ and $p(y \mid u = 20, \theta)$ does not depend on θ . For $u = 0$, $S(0, \theta) = \{\theta/2\}$ and $y \in \{\theta/2 - 1, \theta/2 + 1\}$; if both signs of w occur among the m observations, the two observed values determine $\theta/2$ exactly, which happens with probability $1 - 2^{1-m}$. Both heuristics measure slack in an optimality condition evaluated at the noisy y_i ; at $u = 20$ the values $y = 9$ and $y = 11$ are suboptimal or infeasible by amounts that depend on θ , so these observations enter the loss with θ dependent weight even though they carry no information about θ . Removing them, both criteria are minimized at θ_0 . $\square$

Slack based losses let observations at the boundary vote on θ in proportion to their constraint violation; the likelihood weights them by their information, which is zero when the constraint is active for every θ nearby. The criterion of Aswani et al. [8] gets this right for Gaussian noise because it is the Gaussian likelihood. The paper does not say so, and so misses that under the Rademacher noise of its own counterexample the posterior is exact after finitely many observations while least squares converges at rate $n^{-1/2}$. The general statement is the likelihood principle [12]: data sets with proportional likelihoods carry the same evidence, and a loss that distinguishes them discards or invents information.

2.4. Linear forward problems are change point problems

Take the first synthetic example of Aswani et al. [8], FOP A: $\min\{(\theta + u)x : x \in [-1, 1]\}$ with $u \sim U[-1, 1]$. Then $S(u, \theta) = \{-\text{sign}(\theta + u)\}$ almost surely and

$$y_i = -\text{sign}(\theta + u_i) + w_i.$$

n	$\sigma = 1$			$\sigma = 2$		
	n^2 MSE, LS	n^2 MSE, PM	coverage	n^2 MSE, LS	n^2 MSE, PM	coverage
50	12.0	8.1	0.951	95.8	63.2	0.956
100	12.8	8.3	0.952	125.3	88.1	0.949
200	12.9	8.8	0.951	128.7	89.7	0.949
400	12.0	8.4	0.953	128.8	91.3	0.951
800	12.5	8.4	0.954	113.8	84.2	0.948
1600	12.4	8.1	0.950	121.5	89.6	0.949

Table 1: Inverse linear programming (FOP A of Aswani et al. [8]), $\theta_0 = 0.3$, 3000 replications. LS is the least squares inverse estimator (equal to the Gaussian MLE, midpoint of the argmax interval); PM is the flat prior posterior mean computed exactly from (2). Stability of n^2 MSE confirms the n^{-1} rate. The posterior mean reduces MSE by a factor of about 1.45, and 95 percent equal tail credible intervals have frequentist coverage of 95 percent.

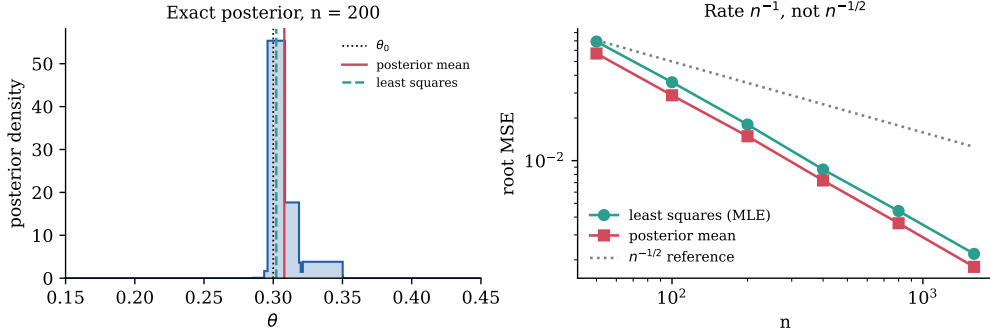


Figure 1: Left: the exact posterior for FOP A with $n = 200$, $\sigma = 1$ is a step function concentrated on an interval of width $O(1/n)$ around $\theta_0 = 0.3$. Right: root MSE of least squares and posterior mean against n ; both decay as n^{-1} .

This is a threshold regression with the threshold at $-\theta$. The log likelihood is a step function of θ with jumps at $\theta = -u_i$; with Gaussian noise of variance σ^2 ,

$$\ell_n(\theta) = \text{const} - \frac{2}{\sigma^2} \sum_{i: -u_i < \theta} y_i, \quad (2)$$

so the posterior under a flat prior is piecewise constant and can be computed exactly in $O(n \log n)$ by sorting the breakpoints. No bilevel program, no ϵ regularization and no δ net are needed.

The theory of discontinuous likelihoods [22, 28, 30, 42, 49] gives three predictions: the estimation error is $O_p(n^{-1})$ rather than $O_p(n^{-1/2})$; the limit law of $n(\hat{\theta} - \theta_0)$ is the argmax of a two sided compound Poisson process, so no Bernstein von Mises theorem holds; and the maximum likelihood estimator is not asymptotically efficient, whereas Bayes estimators under squared loss are. Chernozhukov and Hong [30] prove the last point for a general class of econometric models with jumps in the likelihood. The FOP A errors in Table 1 of Aswani et al. [8] (0.0211 at $n = 100$, 0.0009 at $n = 1000$) already decay faster than $n^{-1/2}$, though the paper does not remark on it.

Table 1 and Figure 1 confirm all three predictions. The posterior is also the only procedure here with a calibrated statement of uncertainty. For strictly convex problems Bayes and least squares agree to first order (Section 2.6); the gains concentrate where constraints bind, which is the generic case for the linear and integer programs of operations research.

2.5. The limit experiment for inverse linear programs

The change point structure can be made precise. Write $h = n(\theta - \theta_0)$ and let $Z_n(h) = \ell_n(\theta_0 + h/n) - \ell_n(\theta_0)$ be the local log likelihood ratio for FOP A.

PROPOSITION 2. *In FOP A with $u \sim U[-1, 1]$, Gaussian noise of variance σ^2 and $\theta_0 \in (-1, 1)$, Z_n converges in distribution, in the Skorokhod topology on compacts, to the two sided compound Poisson process*

$$Z(h) = \sum_{k \leq N_+(h)} J_k^+ \quad (h > 0), \quad Z(h) = \sum_{k \leq N_-(-|h|)} J_k^- \quad (h < 0),$$

where $N_\pm$ are independent Poisson processes of rate $1/2$ and the jumps $J_k^\pm$ are independent $N(-2/\sigma^2, 4/\sigma^2)$. Consequently

$$n(\hat{\theta}_{\text{LS}} - \theta_0) \Rightarrow \operatorname{argmax}_h Z(h), \quad n(\hat{\theta}_{\text{PM}} - \theta_0) \Rightarrow \frac{\int h e^{Z(h)} dh}{\int e^{Z(h)} dh},$$

and the posterior of h converges to the random density proportional to $e^{Z(h)}$.

Proof. By (2), for $h > 0$, $Z_n(h) = -(2/\sigma^2) \sum y_i$ over the i with $\theta_0 < -u_i < \theta_0 + h/n$. The points $-u_i$ have density $1/2$ near θ_0 , so after rescaling by n they form a binomial process converging to a Poisson process of rate $1/2$. For these i we have $\theta_0 + u_i < 0$, hence $x^* = +1$ and $y_i = 1 + w_i$, so each jump is $-(2/\sigma^2)(1 + w_i) \sim N(-2/\sigma^2, 4/\sigma^2)$. For $h < 0$ the relevant i have $x^* = -1$, $y_i = -1 + w_i$, and the jump $(2/\sigma^2)(-1 + w_i)$ has the same law. Independence across sides and of jumps from locations follows from the independence of (u_i, w_i) . The statements about the estimators follow from the argmax continuous mapping theorem and from the theory of Bayes estimators for models with discontinuous densities in Ibragimov and Has'minskii [49]. $\square$

The limit experiment gives exact constants. Simulating Z directly (20,000 paths) gives $\mathbb{E}h_{\text{LS}}^2 = 12.7$ and $\mathbb{E}h_{\text{PM}}^2 = 8.6$ at $\sigma = 1$, and 128.4 and 89.5 at $\sigma = 2$, matching the finite sample n^2 MSE in Table 1 within simulation error. The efficiency loss of least squares, a factor of about 1.45, is a property of the limit and does not vanish with more data. Three remarks follow. First, the Fisher information is infinite, so root n theory, sandwich standard errors and the naive bootstrap are all invalid; the bootstrap is known to fail for change point estimators [76]. Second, the constants grow quickly with σ , by a factor of about ten from $\sigma = 1$ to $\sigma = 2$; for large σ the process Z is close to a Brownian motion with drift $-|h|/\sigma^2$ and variance $2|h|/\sigma^2$, whose argmax scales as σ^2 , so n^2 MSE grows like σ^4 . Third, the posterior has the correct nonstandard shape at finite n , which is why its intervals are calibrated in Table 1 without asymptotic correction.

The limit experiment also settles which estimator to use. Write the limit data as the Poisson points s_k of rate $1/2$ on $\mathbb{R}$ (the rescaled breakpoints $-n(u_i + \theta_0)$) with marks $y_k = \operatorname{sign}(s_k - t) + w_k$, where t is the local parameter. Shifting t and all s_k by the same amount leaves the law of the data unchanged, so the limit is an exact location experiment.

PROPOSITION 3. *In the limit experiment of Proposition 2, the posterior mean under the flat prior, $\int h e^{Z(h)} dh / \int e^{Z(h)} dh$, is the Pitman estimator. It is the minimum risk equivariant estimator under squared loss and is minimax. The least squares estimator $\operatorname{argmax}_h Z(h)$ is equivariant with strictly larger constant risk, so it is not asymptotically minimax: for every $c > 0$,*

$$\liminf_{n \rightarrow \infty} \sup_{|t| \leq c} n^2 \mathbb{E}_{\theta_0 + t/n} (\hat{\theta}_{\text{LS}} - \theta_0 - t/n)^2 > \lim_{n \rightarrow \infty} \sup_{|t| \leq c} n^2 \mathbb{E}_{\theta_0 + t/n} (\hat{\theta}_{\text{PM}} - \theta_0 - t/n)^2.$$

Proof. For a location experiment the generalized Bayes estimator under the flat prior is the best equivariant estimator under squared loss, and it is minimax by the theorem of Girshick and Savage

[44]. The least squares estimator is equivariant, hence has constant risk, and it differs from the Pitman estimator with positive probability, so its risk is strictly larger. The asymptotic statement follows from the convergence of the local experiments to the limit and the local asymptotic minimax theorem for such experiments [49, Chapters I and V]; the constants are 12.7 and 8.6 at $\sigma = 1$. $\square$

The situation is the reverse of the regular case. There the maximum likelihood estimator is asymptotically efficient and Bayes estimators merely match it; here the likelihood maximizer is dominated by a likelihood average. The least squares estimator of Aswani et al. [8] is the wrong functional of the right likelihood.

2.6. The strictly convex case is a regular nonlinear regression

Aswani et al. [8] state that much of the classical theory of least squares is invalid for inverse optimization, and build their results on epiconvergence. For strictly convex forward problems this overstates the difficulty. Under strong second order sufficiency and a linear independence constraint qualification, the solution map $\theta \mapsto x^*(u, \theta)$ is locally Lipschitz and directionally differentiable, and it is differentiable at every (u, θ) where strict complementarity holds [18, 38, 73]. If the set of inputs u at which the active set changes has probability zero, the inverse problem is an ordinary nonlinear regression with a Lipschitz mean function, and the standard theory applies.

PROPOSITION 4. *Suppose f is strongly convex in x , the constraint qualification above holds, $J(u, \theta) = \partial_\theta x^*(u, \theta)$ exists for almost every u at θ_0 , $V = \mathbb{E}[J(u, \theta_0)^\top J(u, \theta_0)]$ is nonsingular, and the noise is $N(0, \sigma^2 I)$. Then the model is differentiable in quadratic mean with Fisher information V/σ^2 , it is locally asymptotically normal, the least squares estimator of Aswani et al. [8] satisfies $\sqrt{n}(\hat{\theta} - \theta_0) \Rightarrow N(0, \sigma^2 V^{-1})$, and the posterior satisfies the Bernstein von Mises theorem. By the local asymptotic minimax theorem no estimator has smaller asymptotic risk uniformly in shrinking neighborhoods of θ_0 .*

Proof. The criterion $m_\theta(u, y) = \|y - x^*(u, \theta)\|^2$ is Lipschitz in θ with square integrable constant, because x^* is Lipschitz and bounded under the regularity condition R1 of Aswani et al. [8], and it is differentiable at θ_0 almost surely. Theorem 5.23 of van der Vaart [84] then gives asymptotic normality with the stated variance. For a Gaussian location model, differentiability in quadratic mean is equivalent to $\|x^*(u, \theta_0 + h) - x^*(u, \theta_0) - Jh\|_{L_2(P_u)} = o(\|h\|)$, which holds by the Lipschitz bound, almost sure differentiability and dominated convergence; the score is $J^\top(y - x^*)/\sigma^2$. This gives local asymptotic normality, Bernstein von Mises (84, Theorem 10.1) and the local asymptotic minimax bound [46]. $\square$

For FOP B of Aswani et al. [8], $x^* = \text{clip}((\theta + u)/2, 0, 1)$, so $J = 1/2$ on the event $\{0 < \theta + u < 2\}$ and 0 otherwise. With $u \sim U[0, 2]$, $\theta_0 = 1/2$ and $\sigma = 1$, this event has probability $3/4$, the Fisher information per observation is $3/16$, and the asymptotic standard deviation of any regular estimator is $\sqrt{16/(3n)}$. The implied asymptotic mean absolute error, $\sqrt{16/(3n)}\sqrt{2/\pi}$, is 0.058 at $n = 1000$.

Table 2 compares this with the paper. Our replication of the stated design is within ten percent of the information bound from $n = 100$ on. The values in Table 1 of Aswani et al. [8] are smaller by a factor of four at $n = 100$ and nine at $n = 1000$, and they decay like $n^{-0.9}$ rather than $n^{-1/2}$. A regular estimator in a locally asymptotically normal model cannot do this, and least squares is regular here by Proposition 4. Nor does any single noise level fit the column: matching $n = 1000$ requires $\sigma \approx 0.11$, matching $n = 100$ requires $\sigma \approx 0.27$. We raise this as a question for the authors. It matters for the comparison with KKA and VIA in the same table, which may have been run in a much less noisy regime than stated, where the difference between consistent and inconsistent estimators is smallest. The FOP A entries, by contrast, are reproducible: with $\theta_0 = 1$ we obtain mean absolute errors of 0.023 and 0.0004 at $n = 100$ and 1000, against 0.021 and 0.0009 reported.

n	reported in Aswani et al. [8], Table 1	our replication of the stated design	asymptotic bound
10	0.4577	0.522	0.583
30	0.2481	0.367	0.336
50	0.1510	0.301	0.261
100	0.0501	0.201	0.184
300	0.0222	0.113	0.106
500	0.0123	0.087	0.082
1000	0.0063	0.057	0.058

Table 2: Mean absolute estimation error of the enumeration (least squares) estimator for FOP B: $u \sim U[0, 2]$, $w \sim N(0, 1)$, $\theta_0 = 1/2$, $\Theta = [0, 2]$, grid spacing 0.01, 400 replications. The last column is the mean absolute error implied by the Fisher information of Proposition 4; it is asymptotic, so finite sample errors may exceed it, and at $n = 10$ the restriction to Θ keeps them below it.

That design places θ_0 on the boundary of Θ and the threshold $-\theta_0$ at the edge of the support of u , a boundary case excluded from Proposition 2 in which the limit process is one sided.

Proposition 4 also bears on the semiparametric algorithm. Under assumption A3 of Aswani et al. [8] the objective is affine in θ and the forward problem is strictly convex, a regular parametric model in which the likelihood or the posterior attains the information bound at rate $n^{-1/2}$. The two step procedure instead denoises each y_i with a Nadaraya Watson smoother and then minimizes suboptimality, so its accuracy is limited by a nonparametric first stage whose only stated bound is $n^{-1/(18m)}$. It gives up a parametric rate the model already provides.

2.7. Information geometry of the solution map

Propositions 2 and 4 are two ends of one picture, in which the rate and the information are properties of the geometry of $u \mapsto x^*(u, \theta)$, not of any estimator. We take the constraints free of θ , as in assumption A3 of Aswani et al. [8], and write L for the Lagrangian.

PROPOSITION 5. *Let $x^*(u, \theta_0)$ satisfy the linear independence constraint qualification, strict complementarity and second order sufficiency, with active constraint set $A(u)$, for almost every u . Let $Z(u)$ be an orthonormal basis of the tangent space $\{d : \nabla g_A(x^*)^\top d = 0\}$, $H(u) = \nabla_{xx}^2 L$ and $B(u) = \nabla_{x\theta}^2 f$ at (x^*, θ_0) . Then*

$$J(u) = \partial_\theta x^*(u, \theta_0) = -Z(Z^\top H Z)^{-1} Z^\top B, \quad I(\theta_0) = \frac{1}{\sigma^2} \mathbb{E}_u \left[B^\top Z (Z^\top H Z)^{-2} Z^\top B \right].$$

Consequently $I(\theta_0)$ is nonsingular, which gives local identification at rate $n^{-1/2}$, if and only if for every direction $\xi \neq 0$, $\Pr_u\{Z(u)^\top B(u)\xi \neq 0\} > 0$.

Proof. Differentiate the KKT system $\nabla_x f + \nabla g_A \lambda_A = 0$, $g_A(x) = 0$ in θ : $H dx + \nabla g_A d\lambda_A + B d\theta = 0$ and $\nabla g_A^\top dx = 0$. The second equation gives $dx = Z dz$; premultiplying the first by $Z^\top$ eliminates $d\lambda_A$ and gives $Z^\top H Z dz = -Z^\top B d\theta$. Strict complementarity keeps the active set locally constant, so this is the derivative [38]. The Fisher information of the Gaussian model is $\mathbb{E}[J^\top J]/\sigma^2$, and $Z^\top Z = I$. Since $(Z^\top H Z)^{-2}$ is positive definite, $\xi^\top I \xi = 0$ exactly when $Z^\top B \xi = 0$ almost surely. $\square$

The formula says that a parameter is seen only through the component of $\nabla_\theta \nabla_x f$ that lies in the tangent space of the active constraints; the normal component is absorbed by the multipliers. This is the mechanism behind the identification failures in Aswani et al. [8]. In FOP B, $H = 2$, $B = -1$, $Z = 1$ in the interior and Z is empty when a bound is active, which gives $J \in \{1/2, 0\}$ as used in Section 2.6. At a vertex of a polytope the tangent space is $\{0\}$, so $J = 0$ and the regular

information vanishes entirely: all information about θ comes from inputs at which the optimal vertex switches. The paper's condition IC is a global, set valued version of the rank condition above; the rank condition is local, checkable, and gives the asymptotic variance.

When the regular information vanishes, the rate is read off the Hellinger geometry. For Gaussian decision noise the Hellinger affinity of $N(a, \sigma^2 I)$ and $N(b, \sigma^2 I)$ is $\exp\{-\|a - b\|^2/(8\sigma^2)\}$, so

$$H^2(P_\theta, P_{\theta_0}) = \mathbb{E}_u \left[1 - \exp \left\{ -\frac{\Delta(u, \theta)}{8\sigma^2} \right\} \right], \quad \Delta(u, \theta) = \|x^*(u, \theta) - x^*(u, \theta_0)\|^2.$$

PROPOSITION 6. *Suppose the feasible sets have diameter at most D . Then $c_D \mathbb{E}_u \Delta/(8\sigma^2) \leq H^2(P_\theta, P_{\theta_0}) \leq \mathbb{E}_u \Delta/(8\sigma^2)$ with $c_D = (1 - e^{-D^2/8\sigma^2})/(D^2/8\sigma^2)$. If $\mathbb{E}_u \Delta(u, \theta) \asymp \|\theta - \theta_0\|^\alpha$ near θ_0 , the minimax rate of estimation is $n^{-1/\alpha}$, and a posterior from a prior with positive continuous density contracts at θ_0 at rate $n^{-1/\alpha}$ up to logarithmic factors. Under the conditions of Proposition 5 with nonsingular $I(\theta_0)$, $\alpha = 2$. For a linear program whose cost is linear in (u, θ) and whose input has a bounded density that is positive near the switching boundary $\{u : x^*(u, \theta_0) \text{ is not unique}\}$, and whose switching boundary moves with nonzero speed in θ , $\alpha = 1$.*

Proof. The bounds follow from $t(1 - e^{-T})/T \leq 1 - e^{-t} \leq t$ for $0 \leq t \leq T = D^2/(8\sigma^2)$. Rates of $n^{-1/\alpha}$ under $H^2 \asymp \|\theta - \theta_0\|^\alpha$ are the general result of Ibragimov and Has'minskii [49, Chapter I], and posterior contraction in Hellinger distance at rate $n^{-1/2}$ up to logarithms, which translates into $n^{-1/\alpha}$ in θ , is Ghosal et al. [43]. In the regular case $\mathbb{E} \Delta = (\theta - \theta_0)^\top \mathbb{E}[J^\top J](\theta - \theta_0) + o(\|\theta - \theta_0\|^2)$. For the linear program, Δ is zero unless the optimal vertex changes, and is then bounded above by D^2 and below by the squared minimum distance between distinct vertices. The vertex changes only for u in a slab around the switching boundary whose width is proportional to $\|\theta - \theta_0\|$, which has probability of that order under the density assumption. $\square$

Two consequences matter for practice. First, the exponent α is a property of the design as well as the model: if the input density vanishes like $d(u)^\beta$ at the switching boundary, where $d(u)$ is the distance to it, then $\alpha = 1 + \beta$ and the rate slows to $n^{-1/(1+\beta)}$. Inputs placed near switching boundaries carry the information, which turns the choice of u in experiments, such as the contract experiment of Section 5, into a Bayesian design problem. Second, under objective space noise the likelihood has an exact polyhedral form. If the cost of a linear program over a polytope P is $c(u, \theta) + \tau\xi$ with $\xi \sim N(0, I)$, vertex v is chosen with probability equal to the Gaussian measure of $c(u, \theta) + \tau\xi \in -N_P(v)$, where $N_P(v)$ is the normal cone at v . As $\tau \rightarrow \infty$ these probabilities tend to the external angles of P , which sum to one over the normal fan [4], and as $\tau \rightarrow 0$ they tend to the indicator of the optimal vertex. The objective space likelihood is therefore a smooth interpolation, indexed by τ , between the change point model of Section 2.4 and a design that carries no information.

2.8. Partial identification

Appendix B of Aswani et al. [8] gives the example FOP II, $\min\{(x - \theta)^2 : x \leq 1\}$ with $\theta \in [0, 2]$, and notes that θ is not identified when $\theta_0 \geq 1$ because every such θ yields $x^* = 1$. The paper imposes identifiability condition IC, and without it accepts convergence of $\hat{\theta}$ to the set of population risk minimizers. Neither says what the data reveal about θ when IC fails, which is common when constraints bind over most of the input range.

PROPOSITION 7. *In FOP II with $\theta_0 \in [1, 2]$, noise $N(0, \sigma^2)$ and a prior π on $[0, 2]$ with a positive density, the posterior converges weakly, in probability, to π restricted to $[1, 2]$ and renormalized. The least squares criterion is minimized by every $\theta \in [1, 2]$, so the least squares estimate is determined by the tie breaking rule of the enumeration.*

Proof. For $\theta \geq 1$ the likelihood is $\prod_i \phi((y_i - 1)/\sigma)$, which does not depend on θ , so the posterior on $[1, 2]$ is proportional to the prior. For $\theta < 1$, $x^* = \theta$, and with $y_i = 1 + w_i$ the log likelihood ratio relative to any $\theta' \in [1, 2]$ is $-n(1 - \theta)^2/(2\sigma^2) + (1 - \theta) \sum_i w_i/\sigma^2$. On $[0, 1 - \delta]$ this tends to $-\infty$ uniformly in probability. On $(1 - \delta, 1)$ it is bounded by its maximum over θ , $(\sum_i w_i)^2/(2n\sigma^2) = O_p(1)$, so that interval carries posterior mass $O_p(\delta)$; let $\delta \rightarrow 0$. $\square$

The posterior thus reports the identified set and, inside it, exactly the information the prior carried, which is the correct answer [45, 62, 68]. Frequentist confidence sets for identified sets exist [31], but they require a different construction from the one in Aswani et al. [8], which offers none. In the paper’s FOP III, $\min\{(x - \theta - u)^2 : x \leq 1\}$ with $\theta \in [0, 2]$, the solution is $\min(\theta + u, 1)$. If the support of u has lower end $\underline{u} > -1$ and $\theta_0 \geq 1 - \underline{u}$, the identified set is $[1 - \underline{u}, 2]$, and the posterior converges to the prior on that interval by the same argument. For inverse linear programs the identified set is the intersection over observed inputs of the normal cones of the observed optimal vertices, which a posterior sampler explores without enumeration.

2.9. The thermostat study is a Tobit regression

The real data application of Aswani et al. [8] fits the forward problem

$$\text{FOP S: } \min_{x \in [70, 76]} \theta_1(x - 76)^2 + (x - \theta_2 - u)^2,$$

where u is the outside temperature and x the setpoint. The first order condition gives the closed form

$$x^*(u, \theta) = \text{clip}(a + bu, 70, 76), \quad b = \frac{1}{1 + \theta_1}, \quad a = \frac{76\theta_1 + \theta_2}{1 + \theta_1}. \quad (3)$$

PROPOSITION 8. *Suppose occupants solve FOP S exactly with a preferred offset $\theta_2 + \eta_i$, $\eta_i \sim N(0, s^2)$ independent of u_i . Then the recorded setpoints follow the two sided Tobit model*

$$y_i = \text{clip}(a + bu_i + e_i, 70, 76), \quad e_i \sim N(0, b^2 s^2),$$

and $(a, b) \mapsto (\theta_1, \theta_2) = (1/b - 1, a/b - 76(1/b - 1))$ is a bijection on $\{0 < b \leq 1\}$. The least squares inverse estimator of Aswani et al. [8] converges to the minimizer (assumed unique) of $\mathbb{E}\{y - \text{clip}(a + bu, 70, 76)\}^2$, which differs from the true (a, b) whenever censoring occurs with positive probability, because $\mathbb{E}[y | u] = \mathbb{E} \text{clip}(a + bu + e, 70, 76) \neq \text{clip}(a + bu, 70, 76)$.

Proof. Replacing θ_2 by $\theta_2 + \eta_i$ in (3) shifts a by $b\eta_i$, which gives the Tobit form. The inverse map follows from $b = 1/(1 + \theta_1)$ and $a(1 + \theta_1) = 76\theta_1 + \theta_2$. The least squares limit is the minimizer of the population criterion by the paper’s own Theorem 3 (FOP S is strictly convex). The conditional mean of a censored normal is a smooth function of u that lies strictly inside $(70, 76)$ and differs from the kinked function $\text{clip}(a + bu)$ on a set of positive probability, so the truth does not minimize the population criterion. $\square$

The grid enumeration over (θ_1, θ_2) is thus replaced by one of the oldest models in econometrics [83] and its data augmentation Gibbs sampler [2, 21, 32]: impute latent uncensored setpoints for observations at 70 or 76 from truncated normals, then draw (a, b) and the scale from their conjugate conditionals. For $n = 1000$ this takes a few seconds and returns the posterior of (θ_1, θ_2, s) .

Table 3 reports the comparison. Least squares converges to the wrong comfort energy tradeoff and its bias does not shrink with n ; the Bayesian Tobit is centered on the truth with close to nominal coverage. Treating censoring at the bounds as noise around a kinked mean, least squares roughly doubles θ_1 , the weight occupants place on saving energy, which is the quantity an energy incentive program would target. If the noise were instead pure measurement error, least squares would be consistent; the point is that the choice should be made by the data.

n	θ_1 (truth 0.5)		θ_2 (truth -10)		BT coverage		reps
	LS	BT	LS	BT	θ_1	θ_2	
100	1.065 (0.381)	0.475 (0.191)	-11.84 (1.20)	-10.00 (0.70)	0.94	0.98	50
300	1.077 (0.183)	0.503 (0.100)	-11.71 (0.65)	-10.04 (0.43)	0.95	0.92	60
1000	1.079 (0.117)	0.499 (0.067)	-11.76 (0.39)	-10.00 (0.25)	0.92	0.92	60

Table 3: Thermostat model FOP S with random utility noise ($\theta_1 = 0.5$, $\theta_2 = -10$, $s = 3$, outside temperature uniform on $[62, 98]$, about 75 percent of setpoints at a bound). LS is the least squares inverse estimator of Aswani et al. [8] computed on a fine grid; BT is the posterior mean from the Bayesian two sided Tobit (Chib data augmentation, flat prior with $\theta_1 \geq 0$, Jeffreys prior on the scale). Entries are means over replications with standard deviations in parentheses; coverage is for 95 percent equal tail credible intervals.

2.10. A hierarchical Bayesian inverse optimization

The constructive alternative has four standard layers:

- observation: $y_i \mid x_i \sim p(y \mid x_i, \sigma)$ (decision space error, if any),
- choice: $x_i \sim \pi_T(\cdot \mid u_i, \theta_i)$ (argmin at $T = 0$, Gibbs or Gumbel at $T > 0$, Section 3),
- heterogeneity: $\theta_i \sim N(\bar{\theta}, \Sigma)$ or a Dirichlet process mixture,
- priors: $\bar{\theta} \sim \pi$, $\sigma, T, \Sigma^{1/2}$ with half Cauchy scales [69].

The heterogeneity layer addresses the paper’s motivation, individualized incentives, as marketing science has for decades: partial pooling of individual preference parameters [3, 74] shrinks noisy individual estimates toward the population and gives a posterior for each individual, so an occupant with forty observations borrows strength from the rest of the building. With $\theta_i = \bar{\theta} + \text{noise}$ and no decision error, the model is the stochastic inverse optimization setting of Birge et al. [16].

Four tools then come for free. Posterior predictive checks test whether the forward model reproduces features of the data, such as the mass of setpoints at the bounds, which FOP S with decision space noise cannot produce. Marginal likelihoods compare forward models; FOP B against FOP C in the paper’s misspecification experiments becomes a Bayes factor rather than a table of prediction errors. Under misspecification the posterior concentrates at the Kullback Leibler projection [55], which is risk consistency stated correctly. And if an analyst insists on a loss rather than a likelihood, the generalized Bayesian update of Bissiri et al. [17] or the quasi posterior of Chernozhukov and Hong [29], $\pi_n(\theta) \propto \exp\{-w n Q_n(\theta)\} \pi(\theta)$, gives uncertainty for the paper’s own criterion; Chernozhukov and Hong [29] designed the quasi posterior precisely for criteria that are discontinuous in θ , which removes the need for the ϵ regularization and the epiconvergence argument.

2.11. Relation to recent work

The literature has begun to move in this direction. Chan et al. [25] propose a hierarchical Bayesian framework with decision space and objective space error, prove posterior consistency by Schwartz’s theorem, and obtain close to nominal coverage; their Appendix also observes that the posterior is step shaped for linear programs. Their sampler is random walk Metropolis that solves N forward problems per iteration, and their reported run times for linear programs reach tens of thousands of seconds at $N = 1000$. Section 2.4 shows that the step shape is the signature of a change point problem with an n^{-1} rate and an inefficient maximum likelihood estimator; exploiting the structure gives the exact posterior in $O(n \log n)$, and for problems without such structure the amortized simulation approach of Section 3 replaces per iteration forward solves by a single training run.

Conformal inverse optimization [56] gives distribution free uncertainty sets for downstream robust optimization but not a model for the data; quantile inverse optimization [78] improves stability of point estimates; contextual and online inverse optimization [15, 86] and inverse optimization with imperfect information [61] remain loss based. A Bayesian treatment unifies these: conformal sets are calibrated predictive sets, quantile losses are asymmetric Laplace likelihoods, and online learning is sequential posterior updating.

2.12. NP hardness

The reduction in Theorem 1 of Aswani et al. [8] uses a forward problem whose solution is $a_\alpha b_\beta c_\gamma$ in closed form. The hardness therefore resides in nonconvex least squares in θ , not in inverting an optimization problem; every sufficiently rich nonlinear regression class contains NP hard instances. Statistically, the hard instances are those with multimodal criteria (here the sign symmetries $(a, b, c) \mapsto (-a, -b, c)$), which is a statement about identifiability. Hardness of computing a point estimate is a reason to report the posterior, which can be explored by Markov chain Monte Carlo or learned by simulation, rather than a reason to restrict to $p \leq 5$.

3. SMOOTHING THE ARGMIN: TEMPERATURE, BOUNDED RATIONALITY AND SIMULATION

Aswani et al. [8] motivate noise as measurement error, bounded rationality, or model misspecification, but model all three as additive noise on x . Bounded rationality has a classical likelihood of its own. Replace the point mass on $S(u, \theta)$ in (1) by the Gibbs measure

$$\pi_T(dx \mid u, \theta) \propto \exp\{-f(x, u, \theta)/T\} \mathbf{1}\{g(x, u, \theta) \leq 0\} dx. \quad (4)$$

As $T \rightarrow 0$, π_T concentrates on $S(u, \theta)$ by the Laplace principle [67], the same fact that underlies simulated annealing [41]; for $T > 0$ it is the multinomial logit of McFadden [59], the quantal response model of McKelvey and Palfrey [60], and the maximum entropy model of inverse reinforcement learning [87]. For a linear program the discrete version is exact: if the cost of each vertex v of the feasible polytope is perturbed by an independent Gumbel shock of scale T , the probability that v is chosen is $\exp\{-f(v, u, \theta)/T\} / \sum_{v'} \exp\{-f(v', u, \theta)/T\}$. The same perturbation device underlies differentiable optimization layers [5, 13], so the machine learning literature has already adopted the smoothing that inverse optimization lacks. Discrete choice likelihoods of this kind are handled by latent variable Markov chain Monte Carlo for discrete regression [2, 21], and logit likelihoods admit exact Gibbs samplers through Pólya Gamma data augmentation [71].

In FOP A, (4) is a truncated exponential on $[-1, 1]$ and

$$\mathbb{E}_T[x \mid u, \theta] = -\mathcal{L}\left(\frac{\theta + u}{T}\right), \quad \mathcal{L}(a) = \coth a - \frac{1}{a},$$

the Langevin function of paramagnetism; the inverse LP becomes a smooth sigmoidal regression in which T measures how far agents are from exact optimality. Three consequences follow. The likelihood is smooth in θ for $T > 0$, so the uniform law of large numbers and Bernstein von Mises apply directly without epiconvergence arguments. The temperature is a parameter with a prior, so the data decide between near exact optimization (the change point regime of Section 2.4) and noisy satisficing, which is the substantive question in most applications. And the ϵ optimal relaxation of Aswani et al. [8], which replaces the argmin by the ϵ sublevel set, is the hard threshold analogue of (4); the Gibbs version has a behavioral interpretation that ϵ lacks.

The Gibbs relaxation also removes the computational hardness that Aswani et al. [8] prove for their formulation. The Gibbs measure (4) is the solution of the variational problem $\min_\pi \{\mathbb{E}_\pi f +$

$T \text{KL}(\pi \| \text{Leb}_\Phi)\}$, its value is the free energy $F_T(u, \theta) = -T \log \int_\Phi e^{-f/T} dx$, and $\nabla_\theta F_T = \mathbb{E}_{\pi_T}[\nabla_\theta f]$, a smooth version of Danskin’s theorem for the value function $V(u, \theta)$.

PROPOSITION 9. *Suppose observed decisions are exact draws from $\pi_T(\cdot | u_i, \theta)$ with T known, and assumption A3 of Aswani et al. [8] holds, $f = f_0 + \sum_{j=1}^p \theta_j f_j$, with the constraints free of θ . Then: (i) the model is an exponential family with sufficient statistic $\phi(x, u) = (f_1, \dots, f_p)(x, u)$ and natural parameter $-\theta/T$, so the log likelihood is concave in θ , the maximum likelihood estimator solves the moment equations $\sum_i \phi(x_i, u_i) = \sum_i \mathbb{E}_{\pi_T(\cdot | u_i, \hat{\theta})} \phi(x, u_i)$, and the Fisher information is $T^{-2} \sum_i \text{Cov}_{\pi_T(\cdot | u_i, \theta)} \phi$; (ii) if in addition f is convex in x for every $\theta \in \Theta$ (assumption A1), each $\pi_T(\cdot | u, \theta)$ is log concave on the convex set $\Phi(u)$, so the gradient of the log likelihood can be estimated to any accuracy by sampling in randomized polynomial time, and under standard boundedness conditions the maximum likelihood estimator, or the posterior mode under a log concave prior, is computable to fixed accuracy in randomized polynomial time; (iii) for f that is L Lipschitz in x on a convex $\Phi(u) \subset \mathbb{R}^d$ with nonempty interior, $0 \leq F_T - V + T \log \text{vol } \Phi \leq LT + dT \log(1/T) + O(T)$ as $T \rightarrow 0$.*

Proof. (i) $\log \pi_T(x | u, \theta) = -\{f_0(x, u) + \theta^\top \phi(x, u)\}/T - \log \int_{\Phi(u)} e^{-(f_0 + \theta^\top \phi)/T} dx$, and the log partition function is convex in θ with gradient $-\mathbb{E} \phi/T$ and Hessian $\text{Cov } \phi/T^2$. (ii) Log concave densities on convex bodies can be sampled, and their integrals approximated, in randomized polynomial time [6, 7, 36, 39, 58]; the score is a difference of an observed and an expected sufficient statistic, so a stochastic gradient method on the concave log likelihood converges at the standard rate. (iii) The lower bound uses $\int e^{-f/T} \leq \text{vol}(\Phi) e^{-V/T}$. For the upper bound, $f \leq V + Lr$ on the ball of radius r about x^* , whose intersection with Φ has volume at least cr^d for small r ; take $r = T$. $\square$

Under the paper’s own assumptions A1 and A3, the inverse problem with exact choices at positive temperature is thus a concave maximum likelihood problem in an exponential family, the maximum entropy inverse reinforcement learning of Ziebart et al. [87] in continuous form, and it is tractable. The hardness reduction of Aswani et al. [8] uses a parametrization that is not affine in θ , but even under A3 their zero temperature criterion with decision noise is nonconvex in θ ; in FOP A, which is affine in θ , it is piecewise constant. The difficulty is a property of that formulation, not of inverse optimization. Part (iii) quantifies the approach to that formulation: the free energy approximates the value function to within $O(dT \log(1/T))$ after a constant shift, so a small temperature changes predicted decisions little while restoring concavity.

When neither (1) nor (4) is tractable, the forward problem is still cheap to solve. That is the setting for simulation based inference [11, 33] and in particular for generative Bayesian computation [70]: draw $\theta^{(j)}$ from the prior and $u^{(j)}$ from the empirical distribution of inputs, solve the forward problem, add noise to obtain $y^{(j)}$, and train a quantile network $\theta = H(y, u, \tau)$ on the simulated pairs. The trained map returns posterior quantiles for any new data set, amortizes the cost over all future data, and never forms the bilevel program. Inverse optimization is close to an ideal case for this approach, because the simulator is a convex solver, and the NP hardness of the point estimation problem is irrelevant to it.

4. FROM ESTIMATES TO DESIGNS

Several papers in the program estimate parameters and then optimize a design with the estimates plugged in. The Medicare Shared Savings Program paper [9] estimates provider behavior with inverse optimization and proposes alternative incentive contracts; the defibrillator paper [26] places devices robustly against uncertainty in cardiac arrest locations; the bike lane paper [57] estimates mode choice and driver routing and then solves a network design program, reporting that 25 miles

of prescribed lanes raise Chicago cycling ridership from 3.6 to 6.1 percent with driving times rising by at most 9.4 percent.

Plug in design has a known failure: the optimized objective evaluated at the estimate is biased upward, the optimizer’s curse of Smith and Winkler [81], because the optimizer selects designs whose estimated value is high partly by chance. The Bayes act maximizes posterior expected utility,

$$d^* = \arg \max_d \int U(d, \theta) p(\theta \mid \text{data}) d\theta,$$

and for the plug in design $\hat{d} = \arg \max_d U(d, \hat{\theta})$ the gap between $U(\hat{d}, \hat{\theta})$ and $\mathbb{E}[U(\hat{d}, \theta) \mid \text{data}]$ measures the curse. For the bike lane study this means reporting a posterior predictive distribution for ridership under the prescribed network, not the single number 6.1 percent, and asking how much of the gain survives integration over uncertainty in the mode choice coefficients and in the congestion response. Two further points are specific to bike lanes. Existing lanes were placed where cyclists already ride, so cross sectional estimates of the lane effect on mode choice are vulnerable to selection; a hierarchical model with prior information from before and after studies of lane openings would discipline it. And driver rerouting can produce Braess type effects, so the posterior over congestion outcomes may be skewed even if the coefficient posterior is not.

For the defibrillator problem the robust formulation guards against the worst case in an uncertainty set whose size is chosen by the analyst. A Bayesian spatial model for arrest intensity, for example a log Gaussian Cox process, delivers the posterior over coverage for every candidate deployment and lets the minimax and Bayes designs be compared on the same predictive scale. Robustness then becomes a choice of loss rather than a choice of set.

For the incentive design problems, the value of information is also computable: the expected improvement in principal utility from one more period of data before committing to a contract. That number tells a regulator whether to redesign now or to experiment first.

5. ESTIMATING INCENTIVE EFFECTS FROM RANDOMIZED CONTRACTS

Kaynar and Siddiq [53] randomize incentive contracts with varying pay rates across Mechanical Turk workers completing the same task and estimate a moral hazard principal agent model, with emphasis on the agent’s production technology, by a consistent estimator expressible as an integer program. The experiment is the strongest part: exogenous variation in contracts is exactly what identifies agent production.

With a finite action set the model is a finite mixture: each worker chooses an unobserved action a , the contract determines the action through incentive compatibility, and the outcome distribution depends on the action. A Bayesian treatment augments each worker with the latent action and samples it jointly with the production parameters, imposing incentive compatibility as a support restriction. Label switching and partial identification, which the integer program resolves by selecting one optimum, appear in the posterior as multiple modes, which is informative. Worker heterogeneity (adverse selection on top of moral hazard) enters as a Dirichlet process mixture over types. Finally, uniform randomization of contracts is not the efficient design once the goal is the optimal contract: Thompson sampling over contracts [47] reaches the optimal contract with far fewer workers, and the posterior from the first wave is the natural prior for the second.

6. GENERATIVE AI, HUMAN CAPITAL AND LABOR COMMODITIZATION

6.1. Design

Siddiq and Zhang [80] assemble a balanced panel of 49,610 Upwork workers over 21 quarters (2021Q1

to 2026Q1, 2.26 million contracts) and ask whether ChatGPT changed what the market rewards. Profile text is embedded with a sentence transformer and grouped into four blocks: self presentation S , credentials C , reputation R and price P . For each quarter q , log demand $y_{iq} = \log(1 + \text{contracts}_{iq})$ is regressed on all blocks by OLS, out of fold; block level Shapley values ϕ_{iq}^k are computed from the 16 subset regressions and satisfy $\hat{y}_{iq} = \bar{y}_q + \sum_k \phi_{iq}^k$. The Shapley values are then outcomes in a continuous treatment difference in differences with occupational exposure E_i from Eloundou et al. [37]. The headline: in the most exposed categories relative to unexposed ones, the three human capital blocks lose 0.0285, 0.0238 and 0.0294 log points of importance (about 7.8 percent combined), price gains 0.0111, and both effects grow by 2025 and 2026. Triple differences show the demand premium of high human capital workers compressing and demand shifting toward lower priced workers in exposed categories. The authors are candid that the aggregate Shapley decline follows mechanically from the demand decline; their argument is that the split across blocks is informative. The paper is careful in several respects: it reports pretrend tests with magnitudes, uses out of fold predictions, checks ridge and XGBoost first stages, and runs a placebo that permutes exposure across subcategories. Our concern is narrower. We show that the split across blocks can also arise mechanically under a natural null, so the design as implemented cannot by itself distinguish commoditization from a level shift in exposed categories.

6.2. What the Shapley design measures

Two features of the design matter. The first stage coefficients are common to all categories within a quarter, so the fitted allocation function cannot differ between exposed and unexposed categories; differential movement in ϕ^k by exposure must come from quarter specific coefficients interacting with category specific covariate distributions. And the embeddings encode category: the authors report that a worker's nearest neighbor in self presentation space shares the broad job category 57.5 percent of the time against 18.6 percent at random, and credentials (employment titles, portfolio projects) and reputation (prior contract titles) are equally occupational.

PROPOSITION 10. *Suppose $y_{iq} = \alpha_{c(i),q} + m(X_i) + \varepsilon_{iq}$, where the allocation m is linear and the same in every quarter, so that clients' valuation of human capital and price never changes, and only the category levels $\alpha_{c,q}$ shift after the shock. Suppose each of the blocks S , C , R contains the category indicators in its linear span and P is uncorrelated with category. Then in the population version of the first stage, for every worker in category c ,*

$$\phi_{iq}^S - \phi_{i,q'}^S = \phi_{iq}^C - \phi_{i,q'}^C = \phi_{iq}^R - \phi_{i,q'}^R = \frac{1}{3}[(\alpha_{c,q} - \bar{\alpha}_q) - (\alpha_{c,q'} - \bar{\alpha}_{q'})], \quad \phi_{iq}^P - \phi_{i,q'}^P = 0.$$

Consequently, if $\alpha_{c,q}$ falls after the shock in proportion to exposure, the difference in differences coefficients on the three human capital blocks are negative and equal, each one third of the demand coefficient, and the price coefficient is zero.

Proof. Write the centered category effect as $\tilde{\alpha}_{iq} = \alpha_{c(i),q} - \bar{\alpha}_q$. Because m is fixed over time, its contribution to every ϕ^k is the same in q and q' and cancels in the differences. The category component defines a cooperative game in which a coalition $\mathcal{S}$ of blocks captures $\tilde{\alpha}_{iq}$ if it contains at least one human capital block and captures nothing otherwise. The marginal contribution of block $k \in \{S, C, R\}$ is therefore $\tilde{\alpha}_{iq}$ when k joins a coalition contained in $\{P\}$ and zero otherwise. With four players the Shapley weight of a coalition of size s not containing k is $s!(3-s)!/4!$, which is $1/4$ for $\emptyset$ and $1/12$ for $\{P\}$, so ϕ^k receives $(1/4 + 1/12)\tilde{\alpha}_{iq} = \tilde{\alpha}_{iq}/3$. Price never changes the captured value and receives zero. The difference in differences regressions are linear in these quantities. $\square$

The symmetry axiom of the Shapley value does the work: blocks that are interchangeable carriers of category information split category level shocks equally. The authors' observation that

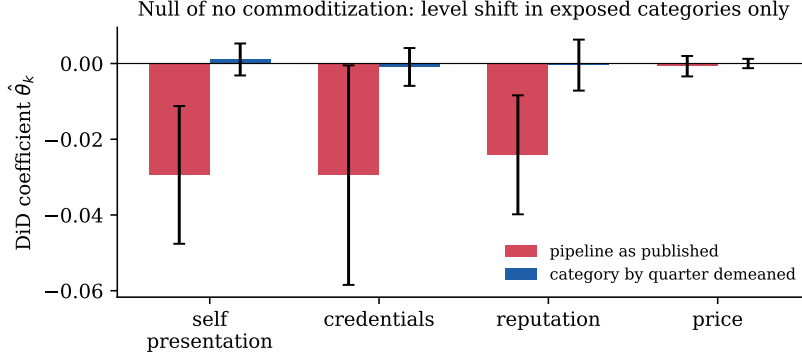


Figure 2: Simulation of the published pipeline under the null of Proposition 10 with imperfect category encoding (12,000 workers, 60 categories, 10 quarters, Poisson contracts, $\log(1 + y)$ outcome, five fold out of fold OLS, exact block Shapley values, pooled continuous difference in differences; 12 replications, bars show two standard deviations). Red bars: the pipeline as published. Blue bars: the same pipeline after demeaning y_{iq} within category and quarter.

the decline is spread almost evenly across the three human capital blocks is exactly what Proposition 10 predicts without any change in how human capital is valued.

Proposition 10 uses perfect encoding. Figure 2 relaxes it: embeddings carry category with noise and also carry latent quality, price depends on category and quality, and demand is Poisson with a level drop of $0.6 E_c$ log points after the shock and no change in the valuation of quality or price. The published pipeline returns human capital coefficients of -0.029 , -0.030 and -0.024 (combined -0.083 against a demand coefficient of -0.21), each at least two replication standard deviations below zero and jointly about five, and a price coefficient of -0.001 . Demeaning y_{iq} within category and quarter before the first stage, which forces the Shapley values to describe the allocation of demand within a category, returns coefficients indistinguishable from zero.

Two conclusions follow. The human capital decline in Siddiq and Zhang [80] is not evidence of commoditization until it survives category by quarter demeaning (equivalently, category by quarter fixed effects as a fifth block that is excluded from the attribution). The price increase is not reproduced by this null, and by the logic of Proposition 10 loading a level shift onto a block correlated with exposure pushes that block’s coefficient down, not up; so the price result is the informative one. It should be read together with the significant price pretrend (joint $p = 0.01$) and with the measurement issue below.

6.3. Measurement and inference

The posted rate and all profile text are from a March 2026 snapshot and are applied to every quarter from 2021. In exposed categories, workers who cut prices to compete after 2022 and who therefore kept demand will appear in the snapshot as low priced, high demand workers in the post period, while their 2021 demand was unrelated to the 2026 price. This dependence of the regressor on post period outcomes can raise the post period importance of price in exactly the categories where the shock was largest. The stability evidence in the appendix of Siddiq and Zhang [80] should be reported separately by exposure, and the analysis repeated on workers whose rates are verified unchanged. The sample also conditions on profiles that remained searchable in 2026; exits that differ by exposure select the post period sample on outcomes.

The outcome $\log(1 + \text{contracts})$ with 52 percent zeros is not scale invariant, so effects in these units depend on the arbitrary constant and mix extensive and intensive margins [27]; a Poisson

or negative binomial model for counts answers the same question with interpretable proportional effects. Inference uses a worker level bootstrap with 100 repetitions, yet treatment varies only across 102 subcategories. With a continuous treatment at the subcategory level, the effective sample for the exposure coefficient is closer to 102 than to 49,610; cluster level resampling or the wild cluster bootstrap [20] is the right default, and the authors’ permutation of exposure across subcategories should be the primary test. In our simulation, with 60 categories, the across replication standard deviation of the demand coefficient is about 0.03; this category level variation is invisible to a worker level bootstrap within one sample. Finally, interpreting θ as the effect of moving from exposure 0 to exposure 1 requires the stronger parallel trends condition for continuous treatments of Callaway et al. [19] as well as linearity in exposure.

6.4. A Bayesian design for the commoditization question

Commoditization is a statement about within category substitution: clients treat workers with different human capital as closer substitutes and respond more to price. That suggests estimating it as such. Let h_i be a scalar human capital index learned only from pre period data (for example a horseshoe regression of pre period demand on the embeddings, with category effects), so that the index is not fit to post period outcomes. Then

$$\text{contracts}_{iq} \sim \text{NegBin}(\mu_{iq}, r), \quad \log \mu_{iq} = \alpha_{c,q} + \gamma_i + \beta_{c,q} h_i + \eta_{c,q} \log p_i,$$

$$\beta_{c,q} = \beta_c + \mathbf{1}\{q \geq q_0\}(\delta_0 + \delta_1 E_c) + \nu_{c,q}, \quad \eta_{c,q} = \eta_c + \mathbf{1}\{q \geq q_0\}(\lambda_0 + \lambda_1 E_c) + \omega_{c,q},$$

with partial pooling of $\beta_c, \eta_c, \nu, \omega$ across the 102 subcategories. Category by quarter effects $\alpha_{c,q}$ absorb level shifts and worker effects γ_i absorb fixed differences (including $\beta_c h_i$ and $\eta_c \log p_i$, so only the post period changes are identified), and commoditization is the joint hypothesis $\delta_1 < 0$ (the human capital gradient flattens) and $\lambda_1 < 0$ (price sensitivity rises) in more exposed categories. The posterior for (δ_1, λ_1) carries category level uncertainty automatically. A variance based importance, the share of within category variance of $\log \mu$ explained by human capital versus price [66], is the magnitude that “importance” should mean; the conditional mean of a signed Shapley value is a decomposition of a level, not a measure of weight.

7. OTHER PAPERS

The ride hailing model of Siddiq and Taylor [79] studies competition between platforms when autonomous vehicles enter. Its comparative statics are derived for given parameter values; a prior predictive analysis, placing distributions on costs and demand elasticities and reporting the probability that each qualitative conclusion holds, would show which conclusions are robust. With commercial robotaxi services now operating in several US cities, the model makes testable predictions about prices and driver earnings that a hierarchical model across cities could confront.

The disintermediation paper [77] treats the accuracy of the signal sellers receive about buyer risk as a platform lever. That lever is information design, and the Bayesian persuasion framework of Kamenica and Gentzkow [52] characterizes the platform optimal signal by concavification of the platform’s value as a function of seller beliefs. Connecting the two would show when partial accuracy is optimal and how commissions and information substitute.

8. SOLUTIONS TO THE OPEN PROBLEMS

The analysis above raises twelve questions. Ten are mathematical or computational and are solved below; for one of them, the optimizer’s curse, the numerical bound needs the authors’ data. The

other two ask for reanalyses of data that are not public; for those we give the method and a simulation of it.

8.1. Decision space or objective space error

PROPOSITION 11. *Let the decision space model have $y = x^*(u, \theta) + w$ with w absolutely continuous on $\mathbb{R}^d$, and let the objective space model return exact optimal decisions of a perturbed problem, with $p_b = \Pr\{y \in \partial\Phi(u)\} > 0$. Then the objective space law is not absolutely continuous with respect to the decision space law: it puts mass p_b on a set that the decision space law gives probability zero. The Bayes factor in favor of the objective space model is infinite as soon as one observation lies on the boundary of the feasible set, and under the objective space model this happens within n observations with probability $1 - (1 - p_b)^n$. If decision noise is added to objective space choices, both models have Lebesgue densities, and the Bayes factor is consistent whenever the models are separated in Kullback Leibler divergence.*

Proof. $\partial\Phi(u)$ is a Lebesgue null set, so it has probability zero under the decision space model and probability p_b under the objective space model; the likelihood ratio on that event is infinite. The laws are mutually singular only when $p_b = 1$, as for linear programs, whose optimal decisions are vertices. The last statement is the standard consistency of Bayes factors between separated parametric models. $\square$

In the thermostat design of Table 3, about 75 percent of setpoints sit exactly at 70 or 76, so a single day of data settles the question that Aswani et al. [8] answer by assumption. Recorded setpoints are typically rounded, which only strengthens the atoms.

8.2. The temperature transition

Consider FOP A with exact choices from the Gibbs measure (4), so that x given u has density proportional to $e^{-(\theta+u)x/T}$ on $[-1, 1]$, and let $T = T_n \rightarrow 0$.

PROPOSITION 12. *The Fisher information per observation is $I_T(\theta_0) = T^{-2}\mathbb{E}_u \mathcal{L}'((\theta_0 + u)/T) = T^{-1}(1 + o(1))$. (i) If $nT_n \rightarrow \infty$, the model is locally asymptotically normal at rate $\sqrt{T_n/n}$, the maximum likelihood estimator satisfies $\sqrt{n/T_n}(\hat{\theta} - \theta_0) \Rightarrow N(0, 1)$, and the Bernstein von Mises theorem holds. (ii) If $nT_n \rightarrow 0$, Lindeberg's condition fails, the information is carried by the $O_p(1)$ observations nearest the threshold, the rate is n^{-1} and the limit is not Gaussian. In the exact vertex case $T = 0$ the posterior is uniform on the interval between the two breakpoints adjacent to θ_0 , and the posterior mean has $n^2 \text{MSE} \rightarrow 2$.*

Proof. The model is an exponential family in θ with log partition $\log\{2\sinh(a)/a\}$, $a = (\theta + u)/T$, so the score is $-\{x + \mathcal{L}(a)\}/T$ and $I_T = T^{-2}\mathbb{E} \mathcal{L}'(a)$. Substituting $s = (\theta_0 + u)/T$, $I_T = (2T)^{-1} \int \mathcal{L}'(s) ds (1 + o(1)) = T^{-1}(1 + o(1))$, since $\mathcal{L}$ increases from -1 to 1 . Each score is bounded by $2/T$, so the Lindeberg ratio is at most $(2/T)/\sqrt{n/T} = 2/\sqrt{nT}$, which vanishes in case (i); the Lindeberg Feller theorem gives local asymptotic normality, and Bernstein von Mises follows for exponential families [84]. In case (ii) the breakpoints $-u_i$ have spacing of order $1/n \gg T$, so the information sum $\sum_i T^{-2} \mathcal{L}'(a_i) \approx \sum_i (\theta_0 + u_i)^{-2}$ is dominated by the nearest breakpoints and is of order n^2 with a nondegenerate random limit. At $T = 0$ the data determine only which breakpoints lie on each side of θ ; the distances from θ_0 to the nearest breakpoint on each side are independent exponentials with rate $n/2$, and the midpoint has variance $2/n^2$. $\square$

Figure 3 confirms the transition at $nT \approx 1$ for n from 200 to 5000. The regimes have a behavioral meaning. If agents optimize with an error of order T in objective units and n observations are

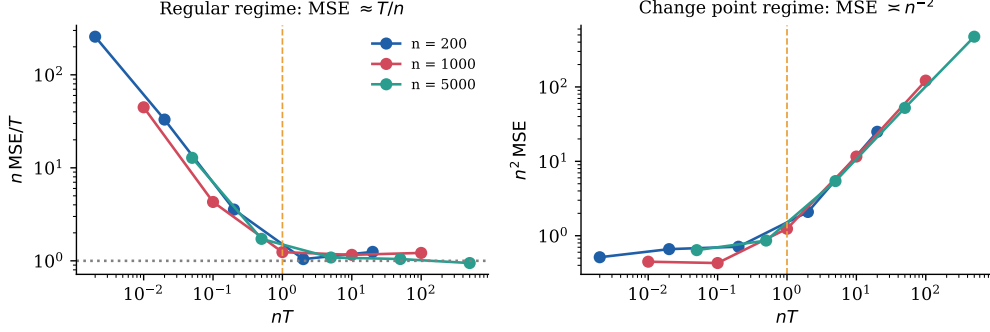


Figure 3: The temperature transition of Proposition 12 for exact Gibbs choices in FOP A (300 replications per point). Left: $n \text{MSE}/T$ is close to one when $nT \gg 1$. Right: $n^2 \text{MSE}$ stabilizes when $nT \ll 1$. The transition is at $nT \approx 1$ (dashed) for every n .

collected, then nT compares the behavioral noise with the spacing of the inputs; inverse optimization is a regular estimation problem when the noise is large compared with the spacing and a change point problem when it is small. In regime (ii), exact recording of decisions adds information beyond the signs: our simulations at $nT \leq 0.1$ give $n^2 \text{MSE}$ near 0.5, below the value 2 for exact vertex data, because a choice at distance d from the threshold deviates from its vertex by an amount of order T/d that reveals d .

8.3. Mixed regimes

PROPOSITION 13. Let $\theta = (\beta, \gamma)$, where β enters x^* through tangent spaces with nonsingular information I_β as in Proposition 5, and γ moves a switching boundary at nonzero speed as in Proposition 6, with the switching set not depending on β to first order. Then with local parameters $h_\beta = \sqrt{n}(\beta - \beta_0)$ and $h_\gamma = n(\gamma - \gamma_0)$,

$$\log \frac{dP_{\beta_0 + h_\beta/\sqrt{n}, \gamma_0 + h_\gamma/n}^n}{dP_{\beta_0, \gamma_0}^n} = h_\beta^\top \Delta_n - \frac{1}{2} h_\beta^\top I_\beta h_\beta + Z_n(h_\gamma) + o_p(1),$$

where $\Delta_n \Rightarrow N(0, I_\beta)$ and $Z_n \Rightarrow Z$, a (multiparameter) compound Poisson process, independent of the limit of Δ_n . The limit experiment is the product of a Gaussian shift and a compound Poisson location experiment. The posterior factorizes asymptotically: Bernstein von Mises for β , the Pitman posterior of Proposition 3 for γ .

Proof. Split the sum over observations into those whose optimal vertex or active set differs between γ_0 and $\gamma_0 + h_\gamma/n$, a set of probability $O(1/n)$, and the rest. On the rest the log likelihood ratio is the regular expansion in h_β ; removing $O_p(1)$ observations changes Δ_n by $O_p(n^{-1/2})$. On the switching set the jumps change by $O(n^{-1/2})$ when β moves by $h_\beta/\sqrt{n}$. Independence follows because Δ_n and the point process are asymptotically determined by disjoint sets of observations. This is the structure found by Chan [22] for threshold autoregression, where slopes are root n normal and the threshold is n consistent and asymptotically independent. $\square$

8.4. Several binding constraints

PROPOSITION 14. Let a linear program have cost linear in (u, θ) , $\theta \in \mathbb{R}^p$, and Gaussian decision noise. Near θ_0 each switching boundary between adjacent vertices v, v' is locally a hyperplane in u

that moves with θ along a normal direction e . Then the local log likelihood ratio in $h = n(\theta - \theta_0)$ converges to

$$Z(h) = \sum_k J_k \mathbf{1}\{s_k \text{ lies between } 0 \text{ and } \langle e_k, h \rangle\}, \quad J_k \sim N\left(-\frac{\|v_k - v'_k\|^2}{2\sigma^2}, \frac{\|v_k - v'_k\|^2}{\sigma^2}\right),$$

where (s_k, e_k, v_k, v'_k) is a Poisson process whose intensity is determined by the density of u on the switching boundaries. The limit is a location experiment in $\mathbb{R}^p$. The flat prior posterior mean is the Pitman estimator, best equivariant and minimax, and least squares is not asymptotically minimax. The rate is n^{-1} in every direction if and only if the normals e_k span $\mathbb{R}^p$ under the intensity.

Proof. An observation contributes to the log likelihood ratio only when its optimal vertex differs at θ_0 and $\theta_0 + h/n$, which happens when its rescaled signed distance s to the switching boundary lies between 0 and $\langle e, h \rangle$. Its contribution is $-\{\|y - v'\|^2 - \|y - v\|^2\}/(2\sigma^2)$ with $y = v + w$, which has the stated normal law; in one dimension $\|v - v'\|^2 = 4$ recovers Proposition 2. The rescaled points converge to a Poisson process by the same binomial argument, with intensity locally uniform in s , so shifting h by t and each s_k by $\langle e_k, t \rangle$ preserves the law. The Pitman and minimax statements are the multivariate versions of Proposition 3 [44]. If the normals fail to span, Z is constant along a direction and the rate in that direction is slower. $\square$

8.5. Profiling over optimal faces

PROPOSITION 15. Let $u \sim U[0, 1]$, $S(u, \theta) = [0, 1]$ for $u < \theta$ and $\{1\}$ for $u \geq \theta$, $\theta \in \Theta = [0, 1]$, and let decisions on a face be uniform, with $y = x + w$, $w \sim N(0, s^2)$. For every θ_0 , the profiled least squares estimator of Aswani et al. [8] converges to $1 = \sup \Theta$. The integrated likelihood, with the uniform selection density on the face, gives a consistent estimator at rate n^{-1} .

Proof. The population criterion is $Q(\theta) = \mathbb{E}[\mathbf{1}\{u < \theta\} d(y, [0, 1])^2 + \mathbf{1}\{u \geq \theta\} (y - 1)^2]$, so $Q'(\theta) = \mathbb{E}[d(y, [0, 1])^2 - (y - 1)^2 \mid u = \theta] < 0$, because $d(y, [0, 1]) \leq |y - 1|$ with strict inequality with positive probability. Hence Q is strictly decreasing and its minimizer is 1, whatever θ_0 . The integrated likelihood has density $\{\Phi(y/s) - \Phi((y - 1)/s)\}$ for $u < \theta$ and $\phi((y - 1)/s)/s$ for $u \geq \theta$; these differ, so the model is identified, and the threshold structure gives the n^{-1} rate as in Proposition 2. $\square$

In simulation (300 replications, $\theta_0 = 0.4$, $s = 0.3$) the profiled estimator averages 0.987, 0.997 and 1.000 at $n = 100, 400$ and 1600, while the posterior mean averages 0.401, 0.400 and 0.400 with mean absolute errors 0.017, 0.004 and 0.0008. The mechanism is general: when solution sets are nested, profiling always prefers the larger set, because a larger set can only bring the projected point closer to y . Integration supplies the Occam factor, the 1/vol normalization of the selection density, that penalizes large sets [51]. Condition IC(iii) of Aswani et al. [8] assumes the problem away; it does not solve it.

8.6. Generative Bayesian computation for integer forward problems

For integer forward problems neither (1) nor (4) has a tractable normalizer, but simulation is easy. We take $x^* = \arg \min_{x \in \{0, \dots, 20\}} \{\theta_1 x^2 - (\theta_2 + u + \eta)x\}$ with preference shocks $\eta \sim N(0, 1)$, $u \sim U[0, 10]$ and $n = 100$ decisions per data set. We simulate 20,000 parameter and data pairs from a uniform prior, reduce each data set to six summaries (regression of x on u , mean, standard deviation, and the fractions at 0 and 20), and fit gradient boosted quantile regressions for the 5, 50 and 95 percent posterior quantiles of each parameter. On 400 new data sets the 90 percent intervals cover θ_1 in 88.0 percent and θ_2 in 85.5 percent of cases, with mean absolute errors of the posterior median of 0.018 and 0.24. As the number of simulations grows and the learner becomes

universal, the learned quantiles converge to the quantiles of the posterior given the summaries [70]; the small undercoverage at 20,000 simulations can be corrected by a conformal recalibration on held out simulations, which we have not run. No bilevel program, enumeration or integer programming solver for the inverse problem is needed.

8.7. Bounding the optimizer's curse

PROPOSITION 16. *Let $\hat{U}_d = U_d + \varepsilon_d$ for designs $d = 1, \dots, K$, with ε centered Gaussian and $\text{Var } \varepsilon_d \leq s^2$, and let $\hat{d} = \arg \max_d \hat{U}_d$. Then*

$$0 \leq \mathbb{E}[\hat{U}_{\hat{d}} - U_{\hat{d}}] \leq s\sqrt{2\log K}.$$

For a network that selects k of m candidate segments, $\log K \leq k \log(em/k)$, so the optimism is at most $s\sqrt{2k \log(em/k)}$, and Dudley's entropy integral sharpens this when networks share segments.

Proof. $\mathbb{E}\hat{U}_{\hat{d}} \geq \mathbb{E}\hat{U}_{d^*} = U_{d^*} \geq \mathbb{E}U_{\hat{d}}$ gives the lower bound. The upper bound is $\mathbb{E}\varepsilon_{\hat{d}} \leq \mathbb{E}\max_d \varepsilon_d \leq s\sqrt{2\log K}$. The bound on K is $\binom{m}{k} \leq (em/k)^k$. $\square$

For the bike lane study the inequality turns the single reported ridership figure into a bracket: with s the posterior standard deviation of predicted ridership for a given network, the reported gain overstates the expected gain by at most $s\sqrt{2k \log(em/k)}$, and the Bayes act removes the overstatement. Evaluating s requires the authors' data.

8.8. When block Shapley values move under the null

PROPOSITION 17. *Let the first stage be population OLS by quarter on time invariant blocks, and let $y_{iq} = g_q(i) + m(X_i) + \varepsilon_{iq}$ with m linear and fixed over time, so that the valuation of every block is constant. Then $\phi_{iq}^k = \phi^k[m](i) + \phi^k[g_q](i)$, where $\phi^k[g_q]$ is the Shapley value of the game $v(\mathcal{S}) = \text{Proj}_{\text{span}(1, X_{\mathcal{S}})} g_q$. The difference in differences coefficient for block k is zero for every k if and only if the change in $\phi^k[g_q]$ from before to after the shock is uncorrelated with exposure. If $g_q(i)$ depends on i only through its category $c(i)$, demeaning y within category and quarter in a balanced panel sets $\phi^k[g_q] = 0$ for every k and quarter, so every coefficient is exactly zero under the null.*

Proof. Projections are linear in the outcome, and the Shapley value is linear in the game, which gives the decomposition; $\phi^k[m]$ is constant over time and cancels. If g_q is constant within categories, demeaning removes it exactly and leaves $m(X_i)$ minus its category mean, which is again time invariant. $\square$

The class of nulls that move block Shapley values is therefore exactly the class of time varying shocks with a component in the span of the blocks that is correlated with exposure. Category level demand shocks are the leading example, and Proposition 10 is the case of perfect encoding. Shocks that vary within categories, for example by worker tenure, require their own controls.

8.9. The commoditization model

The Upwork panel is not public, so we test the model of Section 6 on simulated panels with 6,000 workers in 60 categories over ten quarters, a level shift of $-0.6E_c$ after the shock, and either no change in valuation (the null) or a change of $-0.2E_c$ in the human capital slope and $-0.1E_c$ in the log price slope (commoditization). We estimate (δ_1, λ_1) two ways, both with worker and category by quarter effects: Poisson maximum likelihood on counts, and least squares on $\log(1 + \text{contracts})$.

Table 4 shows that the Poisson model passes the null and recovers both parameters. Least squares on $\log(1 + y)$, with the same fixed effects, does neither: under the null it reports a human

truth	Poisson, counts		OLS, $\log(1 + y)$	
	$\hat{\delta}_1$	$\hat{\lambda}_1$	$\hat{\delta}_1$	$\hat{\lambda}_1$
null, (0, 0)	-0.002 (0.022)	-0.009 (0.035)	-0.056 (0.012)	0.021 (0.015)
commoditization, (-0.2, -0.1)	-0.187 (0.022)	-0.117 (0.044)	-0.111 (0.010)	-0.012 (0.017)

Table 4: Estimates of the commoditization parameters over 15 simulated panels (standard deviations in parentheses). Both estimators include worker and category by quarter effects.

capital decline about five standard deviations from zero, and under commoditization it misses most of the price effect. The reason is the nonlinearity of $\log(1 + y)$ near zero, where a level shift in the mean changes the slope on any regressor [27]. Since 52 percent of worker quarters in Siddiq and Zhang [80] have no contracts, this channel alone could generate part of the reported decline in human capital importance, independently of the Shapley mechanism. Quality signaling predicts $\lambda_1 > 0$ for workers with verified quality, which the same model tests by interacting λ_1 with a verification indicator.

8.10. The thermostat data

The method for the reanalysis is the hierarchical two sided Tobit of Section 2.10, sampled by data augmentation for censored and discrete regression [2, 21, 32]. We simulated 20 occupants with 40 setpoints each, heterogeneous θ_{1j} around 0.5 and θ_{2j} around -10, and the censoring of Table 3. Over 12 replications the root mean squared error of the occupant level posterior medians of θ_{1j} is 0.220 for separate Tobit fits and 0.148 for the hierarchical model, a reduction of one third from partial pooling. Applying this to the Sutardja Dai Hall data requires the data.

8.11. The boundary regime $nT_n \rightarrow \kappa$

Proposition 12 leaves the regime $nT_n \rightarrow \kappa \in (0, \infty)$. It turns out to contain the other two as limits. Write $s_i = n(-u_i - \theta_0)$ for the rescaled breakpoints and $t = n(\theta - \theta_0)$ for the local parameter. Then $(\theta + u_i)/T_n = (t - s_i)/(nT_n)$, so each observation depends on (t, s_i) only through $(t - s_i)/\kappa$.

PROPOSITION 18. *Let $nT_n \rightarrow \kappa \in (0, \infty)$. The local experiments at rate n converge to the experiment E_κ in which one observes the points s_k of a Poisson process of rate $1/2$ on $\mathbb{R}$ and independent marks $x_k \in [-1, 1]$ with density proportional to $\exp\{-(t - s_k)x/\kappa\}$. The log likelihood*

$$\ell_\kappa(t) = \sum_k \left[-\frac{(t - s_k)x_k}{\kappa} - \log \frac{2 \sinh\{(t - s_k)/\kappa\}}{(t - s_k)/\kappa} \right]$$

has increments $\ell_\kappa(t) - \ell_\kappa(t_0)$ that converge almost surely as a series. Moreover:

1. E_κ is a location experiment, so the flat prior posterior mean is the Pitman estimator, best equivariant and minimax, with risk $R(\kappa)$;
2. the Fisher information of E_κ is $1/\kappa$, so $R(\kappa) \geq \kappa$, and $R(\kappa)/\kappa \rightarrow 1$ as $\kappa \rightarrow \infty$, recovering regime (i) of Proposition 12;
3. as $\kappa \rightarrow 0$, the rescaled deviations $M_k = (1 - |x_k|)/\kappa$ converge to independent exponentials with rate $|t - s_k|$, and E_κ converges to an experiment E_0 whose information $\sum_k (t - s_k)^{-2}$ is finite almost surely but has infinite mean. $R(\kappa)$ converges to the Pitman risk c_0 of E_0 . If decisions are recorded only up to their vertex, E_0 reduces to the interval experiment of Proposition 12, with risk 2.

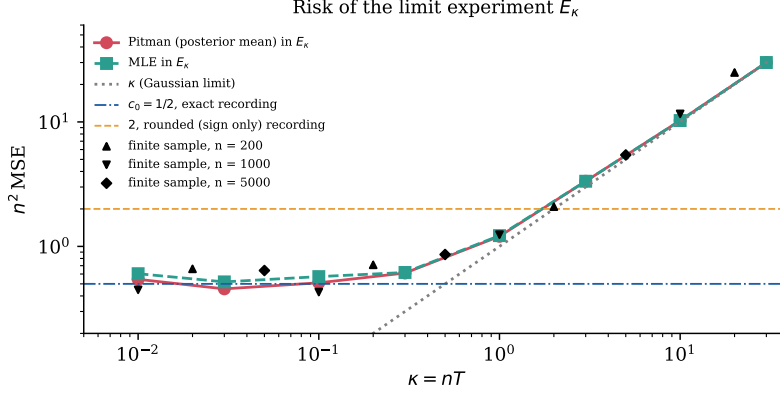


Figure 4: Risk $R(\kappa)$ of the limit experiment E_κ of Proposition 18 (500 replications per point), for the Pitman estimator and the maximum likelihood estimator, with finite sample values of n^2 MSE from Figure 3 (black). The Gaussian bound κ is attained for large κ ; the risk levels off at $c_0 = 1/2$ as $\kappa \rightarrow 0$ with exact recording, and at 2 when only the vertex is recorded.

Proof. Convergence of the rescaled breakpoints to a Poisson process of rate $1/2$ is the binomial argument of Proposition 2, and conditionally on u_i the mark law depends only on $(t - s_i)/(nT_n) \rightarrow (t - s_i)/\kappa$. For a far point with $|s|$ large, $x = \mp 1 \pm Y$ with Y approximately exponential with mean $\kappa/|t_0 - s|$; the increment of its log likelihood between t and t_0 has mean $O(s^{-2})$ and variance $O((t - t_0)^2 s^{-2})$, both summable over a Poisson process, so the series converges by Kolmogorov's three series theorem. Shifting t and every s_k by the same amount preserves the law of the data, which makes E_κ a location experiment; the Pitman and minimax statements follow as in Proposition 3. The Fisher information is $\int \frac{1}{2} \kappa^{-2} \mathcal{L}'((t - s)/\kappa) ds = \kappa^{-1}$, and the Cramér Rao bound for equivariant estimators gives $R(\kappa) \geq \kappa$. As $\kappa \rightarrow \infty$, rescaling t by $\sqrt{\kappa}$ produces many weakly informative points and a central limit theorem for the score, hence a Gaussian shift with information $1/\kappa$. As $\kappa \rightarrow 0$, the truncated exponential with rate $|t - s|/\kappa$ puts its mass within $O(\kappa/|t - s|)$ of a vertex, and $(1 - |x|)/\kappa$ is asymptotically exponential with rate $|t - s|$. The information of an exponential observation about its rate $\lambda = |t - s|$ is λ^{-2} , which gives the stated sum. It is finite because the nearest point is at positive distance, and it has infinite mean because $\int_0^\infty s^{-2} ds$ diverges. Discarding the M_k leaves only the signs, that is the interval experiment. $\square$

Figure 4 computes $R(\kappa)$ by simulating E_κ directly. The risk lies slightly above $\max(\kappa, c_0)$, with $c_0 = 1/2$ to three decimals (Section 8.12). The finite sample values of n^2 MSE from Figure 3 fall on the curve, for example 1.24 at $n = 1000$, $nT = 1$, against $R(1) = 1.21$. The Pitman estimator improves on the maximum likelihood estimator by 10 to 12 percent for $\kappa \leq 0.1$; the two coincide for $\kappa \geq 3$, as they must in the Gaussian limit.

Proposition 18 turns a modeling choice into a number. Under exact optimization with rounded records the best achievable n^2 MSE is 2; recording decisions exactly cuts it to $c_0 = 1/2$; behavioral noise of size T costs nothing until nT approaches c_0 and then raises the risk linearly in nT . The paper's formulation, exact optimization with Gaussian decision noise, is a fourth point on this map (Proposition 2), with risk 8.6 at $\sigma = 1$, more than four times the rounded exact case. Measurement noise blurs the vertex labels that carry the information; behavioral noise below the input spacing does not.

8.12. The constant c_0

The experiment E_0 has more structure than its definition suggests.

PROPOSITION 19. *Let the breakpoints form a Poisson process of rate λ , and let $c_0(\lambda)$ be the Pitman risk of E_0 .*

1. (Scaling) $c_0(\lambda) = c_0(1)/\lambda^2$, and the sign only risk is $1/(2\lambda^2)$.
2. (Sufficiency) *Given the points, let (L, R) be the gap that the signs place t in, and $\beta = \sum_{s_k < t} M_k - \sum_{s_k > t} M_k$. Then β is sufficient for t , and its law is an exponential family in the natural parameter $t \in (L, R)$, with log partition $A(t) = -\sum_k \log |t - s_k|$, mean $\sum_k (t - s_k)^{-1}$ and variance $\sum_k (t - s_k)^{-2}$, the sums taken as principal values. Under the flat prior the posterior is proportional to $|P(t)| e^{-t\beta}$ on (L, R) , where $P(t) = \prod_k (1 - t/s_k)$ is a random entire function whose zeros are the points.*
3. (Moment identity) *The posterior expectation of the mean parameter equals the observation, $\mathbb{E}[\sum_k (t - s_k)^{-1} \mid \text{data}] = \beta$, and the posterior covariance of t and $\sum_k (t - s_k)^{-1}$ is -1 .*

Proof. (1) Multiplying positions by λ and marks by $1/\lambda$ maps E_0 at rate λ to E_0 at rate 1 with t multiplied by λ . The sign only posterior is uniform on the gap containing t , whose length is the size biased version, $\text{Gamma}(2, \lambda)$, of a typical exponential gap G ; hence the risk is $\mathbb{E}[G^3]/(12\mathbb{E}[G]) = 1/(2\lambda^2)$. (2) On (L, R) the order of t and every point is fixed, so $\sum_k |t - s_k| M_k = t\beta + \text{const}$ and the joint density of the marks, $\prod_k |t - s_k| e^{-|t - s_k| M_k}$, has the stated exponential family form. (3) Differentiating the posterior density gives $-\beta - A'(t)$ times the density, and its integral over (L, R) is zero because $|P|$ vanishes at both endpoints; the second identity integrates $(t - \hat{t})$ times the same derivative by parts. $\square$

Proposition 19 reduces E_0 to a one parameter exponential family with a random log partition function, the logarithm of a Poisson random product. We computed $c_0(1/2)$ to high accuracy with this representation. By equivariance the Pitman risk equals the expected posterior variance at $t = 0$, which has much smaller Monte Carlo variance than the squared error. We simulated points within W of the origin and replaced the rest by their exact second order effect, a deterministic term $-t^2/(2W)$ and a Gaussian tilt ξt with $\xi \sim N(0, 1/W)$. With 10^6 replications this gives

$$c_0(1/2) = 0.49986 \text{ (standard error } 0.00071),$$

with agreement within about one standard error at $W = 60$ and on a three times finer grid. We therefore conjecture that $c_0(\lambda) = 1/(8\lambda^2)$, that is $c_0 = 1/2$ at rate $1/2$. Equivalently, exact recording of the deviations from the vertices reduces the risk by exactly a factor of four relative to recording only the vertices. The identity is not a statement about each configuration: conditional on the points, the expected posterior variance is not $G^2/48$. For two isolated points at distance one it is 0.035 rather than $1/48 \approx 0.021$, and it changes with the other points. A proof must therefore average over the Poisson process, presumably through the moment identity (3) and the Palm calculus of the random product P .

8.13. Attribution invariant to the encoding of category

The remaining question from Section 6 is how to attribute demand to blocks so that category level shocks, and the way embeddings happen to encode category, cannot move the attribution. Let D denote the category indicators and, for a set $\mathcal{S}$ of blocks, consider the conditioned projection game

$$v_D(\mathcal{S}) = \text{Proj}_{\text{span}(1, D, X_{\mathcal{S}})} y - \text{Proj}_{\text{span}(1, D)} y,$$

in which category is a player present in every coalition and is not attributed. Let ϕ_D^k be its Shapley values.

PROPOSITION 20. *The attribution ϕ_D has the following properties.*

1. *It is invariant to category level shocks: replacing y by $y + g(D)$ for any function g leaves every ϕ_D^k unchanged.*
2. *It is invariant to how blocks encode category: replacing any block X_k by $X_k + H_k(D)$ for any functions H_k leaves every ϕ_D^k unchanged.*
3. *By the Frisch Waugh Lovell theorem, ϕ_D equals the ordinary Shapley attribution computed after residualizing y and every block on $(1, D)$, which is the category by quarter demeaning of Section 6 applied to both sides.*
4. *It satisfies efficiency, symmetry, the null player property and linearity on the conditioned game, and $\sum_k \phi_D^k = \hat{y} - \text{Proj}_{\text{span}(1, D)} y$, the within category variation of predicted demand.*

The unconditioned attribution of Siddiq and Zhang [80] satisfies neither (1) nor (2) whenever some block's span meets $\text{span}(D)$, by Proposition 10.

Proof. (1) $g(D)$ lies in $\text{span}(1, D) \subseteq \text{span}(1, D, X_S)$ for every S , so it is reproduced by both projections and cancels in $v_D(S)$. (2) $\text{span}(1, D, X_S + H_S(D)) = \text{span}(1, D, X_S)$, so every coalition value is unchanged. (3) The Frisch Waugh Lovell theorem identifies the projection on $\text{span}(1, D, X_S)$ minus the projection on $\text{span}(1, D)$ with the projection of residualized y on residualized X_S . (4) These are the Shapley axioms for the game v_D , and v_D of the grand coalition is the stated difference. The failure of the unconditioned version is the content of Proposition 10. $\square$

Proposition 20 gives the design that the Upwork analysis needs. For any shock that is constant within category and quarter, it returns an attribution that cannot move, by construction. That includes the level shift of Proposition 10 and anything else that is constant within category and quarter, whatever the embedding model. Its sum measures the within category allocation of demand, which is what commoditization is about. The variance based version, the share of within category variance of predicted demand attributed to each block [66], inherits both invariances. Shocks that vary within categories are not removed; for those the model of Section 6 with explicit controls remains necessary.

8.14. What remains open

Two mathematical questions and the data analyses remain. First, a proof that $c_0(\lambda) = 1/(8\lambda^2)$; Proposition 19 supplies the exponential family structure and the moment identity, and the numerical value agrees with $1/2$ to within its standard error of 0.0007. Second, Proposition 20 gives a sufficient construction; whether conditioning on category is the unique Shapley type attribution of projection games satisfying both invariances, in the spirit of the axiomatic characterizations of the Shapley value, is open. Finally, the thermostat, bike lane and Upwork reanalyses need the authors' data; Sections 2.9, 4 and 6 give the methods, and our simulations show what they would deliver.

9. CONCLUSION

The program's central idea, that optimization models are measurement devices for decisions, is a good one, and its statistics would be much stronger if the implied likelihood were written down and used. For constrained problems the likelihood is discontinuous, the rate is n^{-1} and the posterior

beats least squares; for strictly convex problems classical theory applies and gives sharp benchmarks, against which some published results need checking; under failed identification the posterior returns the identified set; bounded rationality calls for objective space error or a temperature, which changes the thermostat estimates; designs should integrate over the posterior rather than plug in; and flexible first stage attributions need a derived null before effects are read as behavior. Each is a modest change to existing code, and each yields uncertainty statements the current papers lack.

Reproducibility. The folder contains `sim_lp_changepoint.py` (Table 1), `sim_cp_limit.py` (Proposition 2), `sim_fopb_cr.py` and `sim_fopa_check.py` (Table 2), `sim_thermostat_tobit.py` and `run_thermo.py` (Table 3), `sim_shapley_null.py` and `mc.py` (Figure 2), and `sim_open_problems.py`, `sim_D_poisson.py`, `sim_hier_tobit.py`, `sim_kappa_limit.py`, `sim_E0.py`, `c0_vec.py` and `c0_conditional.py` (Section 8), and `figs.py` (all figures). Seeds are fixed in the scripts. The null simulation replicates the published pipeline step by step: quarter specific OLS with five fold out of fold prediction, exact block Shapley values from all 16 subset regressions, and the pooled continuous treatment difference in differences of each block’s Shapley value on exposure times post.

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